

> = then click · ▼ = the little red triangle next to that report's title · → = where the answer shows up

SYMBOL KEY — WHAT EACH LETTER MEANS

GREEK = THE TRUE VALUE YOU CAN'T SEE — μ "mew" — the true mean of the whole population · σ "sigma" — the true standard deviation, only when a problem gives it · α "alpha" — the significance level, the cutoff you compare p to (.05, .01) · β "beta" — the chance of missing a real difference · β_0 , β_1 "beta-zero, beta-one" — the true intercept and slope · Σ big sigma — "add up over every data point" · χ^2 "chi-square" — the test JMP prints for a 2x2 table.
LATIN = WHAT YOUR SAMPLE GIVES YOU — $\bar{x}$ "x-bar" — your sample average · s your sample standard deviation · n how many observations · $\hat{p}$ "p-hat" — a sample proportion · b_0 , b_1 the intercept and slope JMP prints · $\hat{y}$ "y-hat" — a predicted y · e a residual: actual — predicted · SE standard error = $\sigma/\sqrt{n}$ or $s/\sqrt{n}$ · r correlation, -1 to +1 · R^2 share of y explained · df degrees of freedom · k how many predictors · t , z , F test statistics. A hypothesis is always about the Greek letter, never $\bar{x}$. Careful: B — alone is the Type II error chance, but B₀ — and B₁ — are regression slopes — same letter, two jobs.

STUDENT CALCULATORS & TEACHING MODULES

GETTING THERE, THEN WHICH FILE — Help > Sample Index → under ① Content Type click Tool → under ② Tool Type pick Calculators → double-click the file. (Some builds list these under the Student menu.) One mean, test statistic + p-value → TestMean.jsl · one proportion → TestProportion.jsl · two means → TestTwoSampleMeans.jsl · two proportions → TestTwoSampleProportions.jsl · CI for a mean → ConfidenceIntervalsForTheMean.jsl · CI for a proportion → ConfidenceIntervalsForProportion.jsl · CI comparing two groups → CITwoMeans.jsl , CITwoProportions.jsl · any normal / t / binomial probability → DistributionCalculator.jsl · how big a sample → SampleSizeForConfidenceIntervals.jsl .
TEACHING MODULES — SHAPE, MEAN, SD, AND A GUIDED TEST — Same path but Tool Type Teaching Modules . Sampling Distribution of Sample Means : set <i>Population Shape</i> and <i>Population Mean</i> → <i>Population Summary</i> gives the shape, Mean and Std Dev, and the sample-means panel gives the mean of $\bar{x}$ and its Std Dev (that's $\sigma/\sqrt{n}$). Hypothesis Test on Mean : pick <i>z-test</i> or <i>t-test</i> , pick the alternative (not equal / less / greater), then tick <i>Reveal Decision</i> .

TESTING ONE MEAN

Is it a number or a group? If you could take an average of it, it's a number → use a **mean**. If it just sorts each person into a group (yes/no, accepted/rejectedd), it's a group → use a **proportion**.
Writing the hypotheses. H_0 always says μ equals the claimed number. "less / went down" → $H_1: \mu < \text{"more / went up"} \rightarrow >$ · "different / changed" → $\neq$.
Shape, average, spread, count. Analyze > Distribution > Y, Columns = variable > OK → the histogram shows the shape; **Summary Statistics** gives Mean, Std Dev, N and Std Err Mean.
t test — when they don't give you σ . Analyze > Distribution > Y = variable > OK > ▼variable > Test Mean > type the claimed number > leave the standard deviation box empty > OK → the **Test Mean** panel, **t Test** row.
z test — when they do give you σ . Same path, but also fill in "Enter True Standard Deviation to do z-test rather than t test" → the panel now shows a **z Test** row instead.
Which p-value to read. H_1 has $\neq$ → **Prob>|t|** · H_1 has $>$ → **Prob>t** · H_1 has $<$ → **Prob<t**. Ignore the Signed-Rank line.
Only summary numbers given. Distribution needs raw data, so use **TestMean.jsl** above, or $t = (\bar{x} - \mu_0) / (s/\sqrt{n})$, $df = n - 1$. If they give s^2 , square-root it first.

WHAT HAS TO BE TRUE, AND INTERVALS

Conditions for the t test. Independence: it's a random sample, it's less than 10% of the whole group, and each person is measured once. **Bell shape:** the problem says bell-shaped, **or** $n \geq 30$ (big samples are fine either way), **or** the histogram looks even with no far-out points.
Checking the shape in JMP. ▼variable > Continuous Fit > Normal > ▼Fitted Normal > Goodness of Fit → **Shapiro-Wilk Prob<W:** if it's bigger than α , the bell shape is fine. Also ▼variable > Normal Quantile Plot — the dots should hug the straight line.
Confidence interval for the mean. **Summary Statistics** already lists Upper / Lower 95% Mean; other levels ▼variable > Confidence Interval > 0.99. Formula $\bar{x} \pm t(\alpha/2, n-1) \cdot s/\sqrt{n}$.

IF H_0 WERE TRUE, AND POWER

Describe the spread of sample averages when H_0 is true. Give all three. **Shape:** bell-shaped, because the population is bell-shaped or $n \geq 30$. **Center:** μ_0 . **Spread:** $\sigma/\sqrt{n}$.
Which sample averages make you reject? cutoff = $\mu_0 + (z)(SE)$, where $SE = \sigma/\sqrt{n}$ and z carries the sign of the tail: -1.645 for a "less than" test, +1.645 for a "greater than" test at $\alpha = .05$; for a two-sided test use $\mu_0 \pm 1.96 \cdot SE$. Reject when $\bar{x}$ is past that cutoff.
 β — the chance of missing a real difference. Take that same cutoff, but measure it from the **true** mean instead: $z = (\text{cutoff} - \mu_{true})/(\sigma/\sqrt{n})$. Then β is the chance $\bar{x}$ lands in the "don't reject" zone.
Power = 1 - β . These all mean power: "chance of detecting this difference", "chance of rejecting H_0 when the true mean is ____", "chance of correctly concluding". These mean β : "chance of failing to reject", "chance of a Type II error".
Power from JMP. DOE > Design Diagnostics > Sample Size and Power > One Sample Mean, fill in Alpha, Std Dev, Difference to Detect and Sample Size, leave **Power** empty > Continue. Leaving **Sample Size** empty instead tells you how big a sample you'd need. It assumes a two-sided test, so on one-sided questions use the formula above.
Drawing it. Two bell curves on one line, one vertical line at the cutoff through both: top curve at μ_0 , tail past the cutoff shaded = α ; bottom curve at the true mean, "don't reject" side shaded = β , the rest power.

TWO PROPORTIONS — SEPARATE GROUPS

A table of counts. File > New > Data Table with Group (Nominal), Outcome (Nominal), Count (Continuous), one row per box, like (Help, Accepted, 222). Then Analyze > Fit Y by X > Y = Outcome > X = Group > Freq = Count > OK → **Contingency Table and Tests**. Skip **Freq** and JMP thinks you have 4 people.
Best guess for the difference. ▼Contingency Table > Row %. Each $\hat{p}$ is the count divided by **that group's own row total**, not the overall total. Example: 222/515 = 55/190 = .4311 — .2895 = **.1416**.
Hypotheses. $H_0: p_1 = p_2 = 0$ — if both groups have the same rate, the difference is zero. "improves" → > 0 · "reduces" → < 0 · "different" → $\neq 0$.
What has to be true. Two separate random groups, or people randomly assigned to them; one answer per person; and all four counts in the table at least 5.
Test and interval. ▼Contingency Analysis > Two Sample Test for Proportions → the difference, its standard error, **Prob>|z|** / **Prob>z** / **Prob<z**, and the confidence interval. **Pearson ChiSquare** in the Tests panel is the same two-sided test.
Formulas. combined $\hat{p} = (x_1+x_2)/(n_1+n_2)$, then $z = (\hat{p}_1-\hat{p}_2) / \sqrt{\hat{p}(1-\hat{p})(1/n_1+1/n_2)}$. The interval does **not** combine them: $(\hat{p}_1-\hat{p}_2) \pm z\alpha/\sqrt{\hat{p}_1\hat{q}_1/n_1 + \hat{p}_2\hat{q}_2/n_2}$.

TWO PROPORTIONS — SAME PEOPLE TWICE

Spotting it and reading the table. Same people measured twice, or matched pairs. **a** and **d** are the people who didn't change; **b** and **c** are the ones who switched, one each way. Smoking: a = 20, b = 5, c = 16, d = 9, n = 50. Don't use the test above — it ignores the pairing.
Only the switchers count. $n' = b + c$. Everyone who stayed the same (a + d) gets dropped. T = how many switched in the direction H_1 expects. Smoking: 29 dropped, $n' = 21$, T = 16. H_0 says switching is equally likely both ways.
The p-value. If H_0 is true, T acts like flipping a coin n' times: T ~ Binomial(n' , 0.5). Cols > New Column > Formula > Discrete Probability > Binomial Distribution(.5, n' , k) gives the chance of k or fewer. Smoking: 1 — BinDist(.5, 21, 15) = **.0133**, under .05, so the video worked.
Or let JMP do it. Analyze > Fit Y by X > Y = After, X = Before, Freq = Count > OK > ▼ > Agreement Statistic → the **test of symmetry** is McNemar's test, but it only gives the two-sided answer.
Interval. $(b-c)/n \pm z\alpha/\sqrt{[(a+d)(b+c) + 4bc] / n^3}$. Smoking: (-.389, -.051), all below zero. Puzzle: (-.196, .077), which includes zero — no real difference.

FILL-IN-THE-BLANK ANSWERS

DEFINE THE PARAMETER — μ = the true mean _____ (with units) for all _____ .
INTERPRET THE TEST STATISTIC —The sample average is _____ standard errors [above / below] _____ , the number H_0 claims.
INTERPRET THE P-VALUE —If the true mean really were _____ , there is a _____ % chance of getting a sample average at least this far from _____ just by chance.
CONCLUSION — WORKS FOR EVERY TEST —Since $p =$ _____ [$<$ / $>$] $\alpha =$ _____ , we [reject / fail to reject] H_0 . There [is / is not] enough evidence to say that _____ . Last blank → ONE MEAN: — the mean [variable] is [less than / greater than / different from] ____ · TWO PROPORTIONS: — the proportion of ____ is [higher / lower / different] for ____ than for ____ · SLOPE: — [x] helps predict [y] · OVERALL F: — the model helps predict [y] · CURVED MODEL: — the squared term makes the model better. Never write "accept H_0 ." If your instructor words it "sufficient evidence to conclude," that is the same sentence.
INTERPRET ANY CONFIDENCE INTERVAL —We are _____ % confident that _____ is between _____ and _____ . Middle blank → M: — the true mean [variable] for all [group] · P₁–P₂: — the difference in the proportion of ____ between ____ and ____ · SLOPE: — the average change in [y] for each extra one [x-unit] · MEAN AT X: — the average [y] for all [units] with $x =$ ____ · PREDICTION: — a single [unit] with $x =$ ____ will have [y]. If the interval includes 0 → no real difference / doesn't help predict.
TYPE I ERROR IN CONTEXT —We decide that _____ when really _____ . The chance of this is $\alpha =$ _____ .
TYPE II ERROR IN CONTEXT —We decide there isn't enough evidence that _____ when really _____ is true. The chance of this is $\beta =$ _____ .
DESCRIBE THE SPREAD OF SAMPLE AVERAGES — Shape: bell-shaped, because _____ . Center: $\mu =$ _____ . Spread: $\sigma/\sqrt{n} =$ _____ / $\sqrt{\text{_____}} =$ _____ . Reason blank: "the problem says the population is bell-shaped" or " $n \geq 30$, so it's fine either way".
CHECK THE T-TEST CONDITIONS — Independence: the _____ are a random sample and fewer than 10% of all _____ . Bell shape: _____ .
POWER AND B —If the true mean is really _____ , there is a _____ % chance we [correctly reject / fail to reject] H_0 . "Correctly reject" = power = $1 - \beta$. "Fail to reject" = β = the chance of a Type II error.
WHICH ERROR WAS POSSIBLE —We [rejected / failed to reject] H_0 , so if we made a mistake it has to be a Type [I / II] error: _____ .
LARGE ENOUGH SAMPLE, TWO PROPORTIONS —The counts _____ , _____ , _____ and _____ are all at least 5, so both groups are big enough.

JMP Exam Notes

CORRELATION AND THE BEST-FIT LINE

Which variable goes where. "Predict A using B" → A is the response and goes in **Y, Response**; B does the predicting and goes in **X, Factor**.

Scatterplot. Analyze > Fit Y by X > Y, Response = response > X, Factor = factor > OK.

Add the line. ▾Bivariate Fit > Fit Line → equation printed right above **Summary of Fit**. For the curved bands around it: ▾Linear Fit > Confid Curves Fit or Confid Curves Indiv (picture only).

Correlation r. Analyze > Multivariate Methods > Multivariate > both variables in Y, Columns > OK → **Correlations**, the number where the two variables cross (the 1.0000s are each variable with itself). Or $r = \pm\sqrt{\text{RSquare}}$, using the sign of the slope. r always lands between −1 and +1: the sign is the direction, the size is the strength — |r| ≥ .8 strong, .5–.8 medium, under .5 weak.

Equation of the best-fit line. Analyze > Fit Model > Y = response > select x > Add > Run → **Parameter Estimates**, **Estimate** column: the Intercept row is b_0 , the x row is b_1 .

Predict. Put the x value into $\hat{y} = b_0 + b_1x$. Watch the units — if x is in thousands, 34,000 means $x = 34$. Don't predict past the range of the data you have.

CHECKING IT'S OK TO USE THE LINE

Save the leftovers. ▾Response y > Save Columns > Residuals, then again for **Predicted Values** and **Studentized Residuals** → new columns at the right of the data table. A residual is just actual – predicted.

Are the leftovers bell-shaped? Analyze > Distribution > Y = Residual > OK > ▾Residual > Normal Quantile Plot, then Continuous Fit > Normal > ▾Fitted Normal > Goodness of Fit → **Shapiro-Wilk**.

Leftovers plot. ▾Response y > Row Diagnostics > Plot Residual by Predicted. A shapeless band around zero is what you want · an arch or U means a line is the wrong shape · a funnel means the spread isn't even · a studentized value past ±3 is a far-out point.

DOES X ACTUALLY HELP PREDICT Y?

Test the slope. Fit Model > Run > Parameter Estimates, on the x row → **t Ratio** is the test number, **Prob>|t|** is the two-sided p-value, **Std Error** is the slope's wobble. H_0 : the slope is 0, $df = n - 2$.

One-sided? Cut **Prob>|t|** in half, but only if the slope's sign matches what H_1 says. If it doesn't, the p-value is over .5 and you fail to reject.

Testing the slope against a number other than 0. JMP only tests 0. Either check whether the confidence interval contains that number, or use $t = (b_1 - c) / \text{Std Error}$, $df = n - 2$.

How much of y is explained. **Summary of Fit** → **RSquare**, times 100 for a percent.

THE THREE INTERVALS

Interval for the slope. Right-click inside **Parameter Estimates** > Columns > Lower 95% + Upper 95% → read them on the **x** row, not the Intercept row. If it doesn't include 0, x helps predict y.

99% or 98% instead. ▾Response y > Set Alpha Level > 0.01 (or **Other** → 0.02), then read those columns again. If yours is stuck at 95%: $b_1 \pm t(\alpha/2, n-2) \cdot \text{Std Error}$, with t from t Quantile(1− α /2, n−2).

Interval for the average y — "for ALL cases with $x = __$ ". Add a row with that x and **leave Y empty** (that keeps it out of the line), then Fit Model > Run > ▾Response y > Save Columns > Mean Confidence Interval → **Lower / Upper 95% Mean** on that row.

Interval for ONE new case. Same setup, then Save Columns > Indiv Confidence Interval → **Lower / Upper 95% Indiv**, always wider. A singular word — "a truck", "a school" — means use Indiv.

Far-out points. Click the point to light up its row, then Rows > Exclude/Unexclude and Run again. "How many more votes than predicted" = actual – predicted, which is that row's residual.

SEVERAL PREDICTORS AT ONCE

Look at them first. Analyze > Multivariate Methods > Multivariate > response + all predictors in Y, Columns > OK → **Correlations** and a grid of mini scatterplots. Two predictors with |r| above about .8 measure nearly the same thing, which causes trouble.

Fit it. Analyze > Fit Model > Y = response > highlight all the predictors > Add > Run → **Parameter Estimates** gives $\hat{y} = b_0 + b_1x_1 + \dots + b_kx_k$.

Does the whole model help? Analysis of Variance → F Ratio and Prob > F. H_0 says every slope is 0. Rejecting means at least one predictor helps — not that they all do.

Does one predictor help, with the others in? Parameter Estimates, that predictor's row → **t Ratio**, **Prob>|t|**, $df = n - k - 1$. A predictor can look strong alone and still be useless here, because another one already covers it.

Interval for one coefficient. Right-click **Parameter Estimates** > Columns > Lower 95% + Upper 95%; change the level first with **Set Alpha Level**.

Checking conditions. Save **Residuals**, use ▾Response y > Row Diagnostics > Plot Residual by Predicted, and run Distribution on the residuals for the plot and Shapiro-Wilk. For overlap between predictors: right-click **Parameter Estimates** > Columns > VIF — over 5 a warning, over 10 a problem.

PICKING THE BEST MODEL

Use adjusted R². **Summary of Fit** → **RSquare Adj**, highest one wins. Plain R² always goes up when you add a predictor, so it can't compare models with different numbers of predictors; $R^2_{\text{adj}} = 1 - (1 - R^2)(n-1)/(n-k-1)$ takes a penalty for extras.

Every model at once. Fit Model > add all > Personality = Stepwise > Run > ▾ > All Possible Models → every combination with RSquare, RSquare Adj, Cp, AICc, BIC.

Dropping predictors one at a time. Fit Model > add all > Stepwise > Run, set Direction = **Backward**, Stopping Rule = **P-value Threshold**, Prob to Leave = **0.1**, click **Go** → **Current Estimates**, a check under **Entered** means it stayed. Then **Make Model** → **Run**. The idea: throw out the biggest p-value above 0.1, refit, repeat.

Predicting from the model you picked. Put the values into that model's equation, or add a row with those x's and an empty Y and save **Predicted Values**, or use ▾Response y > Factor Profiling > Profiler.

CORRELATION · ONE PREDICTOR · SEVERAL PREDICTORS · CURVED

CURVED (QUADRATIC) MODELS

Quick way. Analyze > Fit Y by X > Y, X > OK > ▾Bivariate Fit > Fit Polynomial > 2, quadratic → a **Polynomial Fit Degree=2** block with the plain equation printed above it.

Full way (use this one for intervals). Analyze > Fit Model > Y > select x > Macros > Polynomial to Degree with Degree = 2 > Run → **Parameter Estimates** gains an **x*x** row.

JMP shifts x first. By default it prints $(x-\bar{x}) \cdot (x-\bar{x})$ instead of x^2 , so the numbers aren't the plain b_0, b_1, b_2 . Uncheck ▾Response y > Center Polynomials and Run again to get the plain equation.

Does the curve help? Test the squared term only: **Parameter Estimates**, the **x*x** row (or **Effect Tests**) → **Prob>|t|**. H_0 says the squared term is 0, meaning a straight line is enough. Don't use the overall F here — it can look good just from the straight-line part.

The change in y isn't the same every time. change = $b_1(x_2 - x_1) + b_2(x_2^2 - x_1^2)$. Going 0 to 1 gives $b_1 + b_2$; going 2 to 3 gives $b_1 + 5b_2$. They're different on purpose — that's the whole point of asking both.

Intervals from a curved model. Add the row with x and an empty Y, fit with **Polynomial to Degree 2**, then **Save Columns** → **Mean Confidence Interval** for the average of all such cases, **Indiv Confidence Interval** for one specific new case.

NUMBERS YOU MAY NEED TO LOOK UP

df: $n - 1$ for one mean · $n - 2$ for a slope · $n - k - 1$ for one predictor out of k · k and $n - k - 1$ for the overall F.

z cutoffs: one-sided $\alpha = .10 \rightarrow 1.282, .05 \rightarrow 1.645, .025 \rightarrow 1.960, .01 \rightarrow 2.326$ · two-sided 90% $\rightarrow 1.645, 95\% \rightarrow 1.960, 98\% \rightarrow 2.326, 99\% \rightarrow 2.576$.

THE FORMULAS BEHIND THE LINE

Three building blocks. JMP never shows these three — they're only for working a small data set out yourself. $S_{xx} = \sum (x-\bar{x})^2$ spread of the x's, feeds the slope and its standard error · $S_{yy} = \sum (y-\bar{y})^2$ spread of the y's, feeds r and R^2 · $S_{xy} = \sum (x-\bar{x})(y-\bar{y})$ how much x and y move together, feeds the slope and r.

Everything else comes from those three. slope $b_1 = S_{xy}/S_{xx}$ · intercept $b_0 = \bar{y} - b_1\bar{x}$ (needs b_1 first) · the line $\hat{y} = b_0 + b_1x$ · $r = S_{xy}/\sqrt{(S_{xx} \cdot S_{yy})}$ · $R^2 = r^2$ · one point's leftover $e = y - \hat{y}$. To check your work, your b_0 and b_1 should match **Parameter Estimates**.

FILL-IN-THE-BLANK ANSWERS — REGRESSION

COMMENT ON THE SCATTERPLOT —There is a **[positive / negative]**, **[weak / medium / strong]**, **[straight / curved]** relationship between _____ and _____.

Add if needed: "There is an odd point at ____." If it curves, say a squared term should be tried.

INTERPRET R — $r = ______$ means a **[weak / medium / strong]** **[positive / negative]** straight-line relationship between _____ and _____.

WRITE THE EQUATION —Predicted _____ = _____ + _____ (_____).

INTERPRET THE SLOPE —For every extra one _____, the predicted (average) _____ **[goes up / goes down]** by _____.

INTERPRET THE INTERCEPT —When _____ is 0, the predicted _____ is _____.

If $x = 0$ is outside the data: add "but $x = 0$ isn't in the range of the data, so this doesn't really mean anything."

INTERPRET R² — _____ % of the changes in _____ are explained by the **[straight / curved]** relationship with _____.

DOES X HELP PREDICT Y? —Since $p = ______ [< / >] \alpha = ______$, we **[reject / fail to reject]** H_0 : the slope is 0. There **[is / is not]** enough evidence to say that _____ helps predict _____.

INTERVAL FOR THE SLOPE —We are _____ % confident that for every extra one _____, the average _____ changes by between _____ and _____.

INTERVAL FOR THE AVERAGE Y —We are _____ % confident that the average _____ for all _____ with _____ = _____ is between _____ and _____.

INTERVAL FOR ONE NEW CASE —We are _____ % confident that a single _____ with _____ = _____ will have a _____ between _____ and _____.

WHAT THE LEFTOVERS PLOT SHOWS —The plot shows _____, so the **[straight-line / even spread]** condition **[is / is not]** met. Shapiro-Wilk gives $p = ______ [> / <] \alpha$, so the bell shape **[is / is not]** reasonable. First blank: "a shapeless band around zero" · "an arched pattern" · "a funnel shape".

INTERPRET A COEFFICIENT (SEVERAL PREDICTORS) —For every extra one _____, the predicted _____ **[goes up / goes down]** by _____, while the other predictors stay the same.

DOES THE WHOLE MODEL HELP? —Since $p = ______ [< / >] \alpha = ______$, we **[reject / fail to reject]** H_0 : every slope is 0. There **[is / is not]** enough evidence to say that the model helps predict _____.

ONE PREDICTOR, OTHERS ALREADY IN —With the other predictors already in the model, there **[is / is not]** enough evidence to say that _____ helps predict _____ ($t = ______$, $p = ______$).

PICKING A MODEL —The model with _____ has the highest adjusted R² (_____), so it's the one to use.

DOES THE CURVE HELP? —The squared term has $p = ______ [< / >] \alpha = ______$, so there **[is / is not]** enough evidence to say that the curved model fits better than a straight line.